Agent Capability Graph: Building an RDF Knowledge Graph for Discovering MCP Servers and Agent Skills

Tracks: Select the programs the project is aimed at

Data Science
Internet of Things
Digital Security
Intelligent Communication System
Embedded Systems
Security in Computer Systems and Communications

Name of Main Supervisor

Raphael Troncy

Name of Secondary supervisor (optional)

Thibault Ehrhart

Number of groups possible for this project (MAX 3)

1

Maximum number of students PER GROUP

2

Name of student/s already assigned to this project (optional)

Description for a 100 hours project

Context, Objective and Tasks

Modern LLM-based agents increasingly rely on external capabilities: MCP servers give agents access to tools, APIs, files, databases and services, while Agent Skills package reusable procedural knowledge, scripts and resources that can be loaded when a task requires them. However, the rapid growth of MCP server directories and Skill marketplaces creates a new problem: how can an agent discover the right capability for a task without being overwhelmed by thousands of noisy, duplicated, poorly described or unsafe options?

Most current directories rely on keyword search, categories, tags, README files or vector similarity. These representations are useful, but they do not explicitly model what a server or skill can do, what inputs it requires, what resources it accesses, whether it changes state, what risks it carries, or how it relates to other capabilities. This project explores whether a semantic Knowledge Graph can provide a more structured, explainable and interoperable layer for capability discovery in agentic systems.

The goal of this semester project is to build a prototype RDF Knowledge Graph describing a curated collection of MCP servers and Agent Skills, with a focus on their capabilities. The project will start with a short inventory phase to identify the most relevant MCP server directories and Agent Skill listings to harvest. The students will then design or reuse suitable ontologies to represent servers, tools, skills, parameters, resources, provenance, categories, risks and quality signals.

The overarching research question is:

> **Can a semantic capability graph improve the discovery and selection of MCP servers and Agent Skills compared with ordinary keyword, tag-based or embedding-based search?**

1. **Inventory and selection of sources**
   Identify a curated set of MCP server directories and Agent Skill repositories or marketplaces worth harvesting. The goal is not to collect everything, but to select a meaningful and high-quality subset that covers diverse domains such as software engineering, databases, productivity, web search, document processing, data analysis and knowledge graphs.

2. **Metadata extraction and normalization**
   Extract structured metadata from MCP server listings, tool schemas, README files and Skill descriptions. For MCP servers, this may include server name, tool names, input schemas, resources, prompts, transport, installation method, repository, license and authentication requirements. For Skills, this may include SKILL.md metadata, description, trigger conditions, bundled scripts, reference files, dependencies and supported agent platforms.

3. **Ontology and schema design**
   Study existing vocabularies such as DCAT, PROV-O, DOAP, SKOS and SHACL, as well as earlier work on Semantic Web service descriptions. Reuse existing vocabularies when possible, and design a small domain ontology when needed. The ontology should represent at least: Server, Tool, AgentSkill, Capability, Parameter, Resource, Prompt, SoftwareProject, DirectoryListing, RiskSignal and QualitySignal.

4. **RDF Knowledge Graph construction**
   Build a pipeline that converts harvested metadata into RDF, validates the graph using SHACL, and stores it in a triplestore or RDF-compatible graph database. The graph should preserve provenance, so that each claim can be traced back to the original directory, repository or Skill file.

5. **Capability discovery prototype**
   Implement a small discovery interface that allows users or agents to ask capability-oriented questions such as:
   “Which tools can query a SPARQL endpoint?”,
   “Which Skills help with spreadsheet manipulation?”,
   “Which MCP servers expose file-system operations and require local access?”,
   “Which capabilities are similar but differ in their risks or dependencies?”

6. **Evaluation**
   Compare KG-based discovery with simpler baselines such as keyword search, tag search and embedding search. The evaluation can reuse ideas from SkillsBench and MCP-Atlas: given a set of user tasks, can the system retrieve the relevant MCP server, tool or Skill? The evaluation should report precision, recall, top-k accuracy, and qualitative failure cases.

By the end of the project, the students should deliver:

- A short survey of MCP server directories, Agent Skill listings and related academic work.
- A curated dataset of MCP servers and Agent Skills.
- An RDF ontology or schema, with documentation and examples.
- A populated Knowledge Graph with provenance.
- SHACL validation shapes.
- A prototype search or discovery interface.
- An evaluation comparing KG-based discovery against non-semantic baselines.
- A final report discussing the benefits, limitations and future extensions of semantic capability discovery.

Students should look at SkillsBench as a benchmark for evaluating Agent Skills, MCP-Atlas as a benchmark for tool discovery over real MCP servers, recent empirical studies of MCP and Agent Skill ecosystems, and earlier Semantic Web work on service discovery such as OWL-S, WSMO and SAWSDL. They should also study recent work on representing agentic tools in knowledge graphs, since this project aims to extend this direction toward a unified, RDF-based representation of both MCP servers and Agent Skills.

What you will learn?

This project is ideal for students interested in Knowledge Graphs, Semantic Web technologies, LLM agents, tool use, ontology engineering and information retrieval. Students will gain hands-on experience with RDF, ontology reuse, SHACL validation, metadata extraction, agent capability modeling and evaluation of discovery systems.

Students should be comfortable with Python and data processing. Prior exposure to RDF, SPARQL, Knowledge Graphs or LLM agents is useful but not mandatory. Curiosity about how future AI agents will discover and select tools is essential.

References

[1] SkillsBench: Benchmarking How Well Agent Skills Work Across Diverse Tasks. arXiv:2602.12670, 2026.
https://arxiv.org/abs/2602.12670 | https://www.skillsbench.ai/skillsbench.pdf

[2] MCPZoo: A Large-Scale Dataset of Runnable Model Context Protocol Servers for AI Agent. arXiv:2512.15144, 2025.
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[3] MCP-Atlas: A Large-Scale Benchmark for Tool-Use Competency with Real MCP Servers. arXiv:2602.00933, 2026.
https://arxiv.org/abs/2602.00933

[4] A Measurement Study of Model Context Protocol Ecosystem. arXiv:2509.25292, 2025.
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[5] Model Context Protocol specification — Tools. Model Context Protocol documentation, 2026.
https://github.com/modelcontextprotocol/modelcontextprotocol/blob/main/docs/specification/2026-07-28/server/tools.mdx

[6] Agent Skills: A Data-Driven Analysis of Claude Skills for Extending Large Language Model Functionality. arXiv:2602.08004, 2026.
https://arxiv.org/abs/2602.08004

[7] Agent Skills for Large Language Models: Architecture, Acquisition, Security, and the Path Forward. arXiv:2602.12430, 2026.
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[9] From Anatomy to Smells: An Empirical Study of SKILL.md in Agent Skills. arXiv:2607.01456, 2026.
https://arxiv.org/abs/2607.01456

[10] Agent-as-a-Graph: Knowledge Graph-Based Tool and Agent Retrieval for LLM Multi-Agent Systems. arXiv:2511.18194, 2025.
https://arxiv.org/abs/2511.18194

[11] Representing Agentic Tools in Knowledge Graphs for Structure-Aware Tool Discovery Under Tool Overload. OpenReview preprint, 2026.
https://openreview.net/forum?id=7MVoH9Y3mi

Description for a 180-200 hours project

If time allows, students could go beyond the static Knowledge Graph and build a small agentic demonstrator that uses the graph at runtime to recommend, rank, or compose MCP servers and Skills for a given user task. This extension could also include simple quality and risk signals, such as maintenance status, required permissions, security-sensitive actions, or redundancy between similar capabilities.

In collaboration with (ex: industry partner, etc.)

Tags